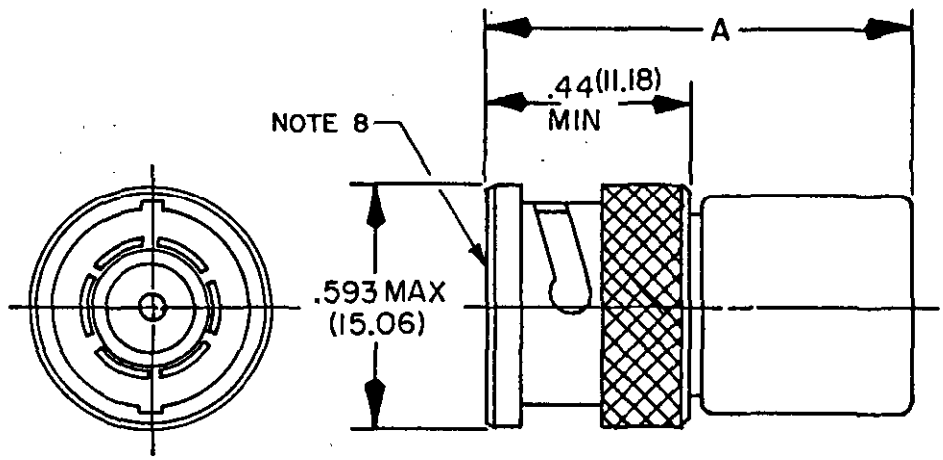
MILITARY SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY, (SERIES BNC (CABLED), PIN CONTACT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.



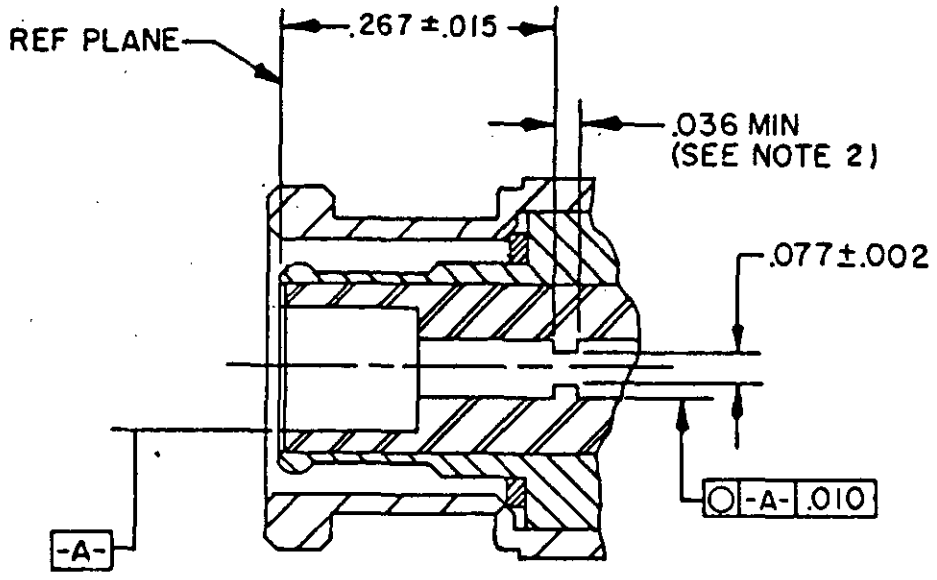
MARKING IMPLEMENTATION DATE, CATEGORY B, SEE TABLE IV

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. For dimension A, see table I and III.
4. Dimension .593 (15.06 mm) is the largest overall diameter of the connector.
5. Wrench flats are to accommodate standard wrench openings in accordance with FED-STD-H28, appendix 10.
6. All undimensioned pictorial representations are for reference purposes only.
7. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
8. Series BNC, pin contact interface in accordance with MIL-STD-348.
9. Metric equivalents are in parentheses.

FIGURE 1. General configuration.

AMSC N/A
FSC 5935

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CATEGORY D

Inches	mm
.002	.05
.010	.25
.015	.38
.036	.91
.077	1.96
.267	6.78

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. In the mated condition the longitudinal force of the spring of the coupling mechanism shall exceed the pressure exerted by the sealing gasket by an amount necessary to insure butting of the outer contacts at the reference plane.

FIGURE 2. Category D captivation detail.

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TABLE I. Dash numbers, cross reference, and dimensions.

Dash number 1/	Applicable cable M17/ #	Dimensions	Inches (millimeters) maximum
Category A - Field serviceable (no special tools required)			
0101 4/	028-RG058 084-RG223* 128-RG400 060-RG142@	A	1.250 (31.75)
0102 4/	029-RG59*Δ 030-RG062Δ 90-RG71Δ 97-RG210Δ		
0103 4/	054-RG122*		
0111 4/	111-RG303@*		
0118 4/	110-RG302*@Δ		
0220 4/	113-RG316@* 119-RG174		
0225 4/	074-RG213*		
Category C - Field replaceable (MIL-C-22520 crimp tool) 3/ See footnote symbol next to applicable cable for crimp die			
0013 4/	028-RG058*~ 111-RG303~	A	1.500 (38.10)
0014 4/	084-RG223*~ 060-RG142~		
0015 4/	29-RG59*Δ> 030-RG062Δ> 97-RG210Δ>		
0016 4/	054-RG122*&		
0017 4/	90-RG71*Δ>		
0020 4/	110-RG302*@Δ>		
0222 4/	113-RG316*Σ 119-RG174Σ		
Category D - Field replaceable - defined piece parts 3/5/2/			
0501 4/	054-RG122*	A	1.500 (38.10)
0502 4/	095-RG180*Δ		
0503 4/	060-RG142* 128-RG400		
0504 4/	028-RG058*		

See footnotes on following page.

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1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ Not to be used in Army equipment.

3/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 4.

4/ These connectors have captivated center contacts.

5/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and ∆.

@ Cable to be used for the +200°C temperature cycling tests.

∆ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

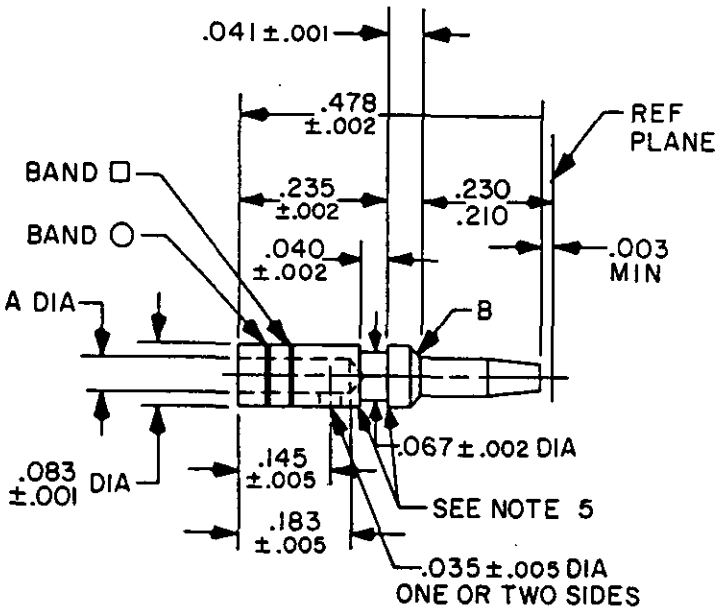
Σ M22520/5-35 Closure B or M22520/5-03 Closure A

& M22520/5-41 Closure B or M22520/5-05 Closure B
M22520/5-09 Closure A

~ M22520/5-19 Closure B or M22520/5-05 Closure A
M22520/5-11 Closure A
M22520/5-57 Closure A

> M22520/5-19 Closure A or M22520/5-07 Closure A
M22520/5-13 Closure A
M22520/5-59 Closure A

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CENTER CONTACT

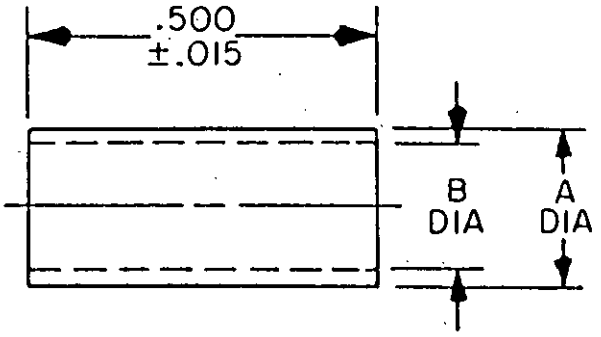
Inches	mm	Inches	mm
.001	.03	.145	3.68
.002	.05	.175	4.45
.003	.08	.183	4.65
.005	.13	.206	5.23
.015	.38	.210	5.33
.017	.43	.212	5.38
.033	.84	.220	5.59
.035	.89	.230	5.84
.040	1.02	.235	5.97
.043	1.09	.250	6.35
.067	1.70	.478	12.14
.083	2.11	.500	12.70

Dash No.	Contact No. 2/	A +0.001 -0.002	B	Basic crimp tool 1/	Crimp die or positioner	Crimp tensile min	Color band □	Color band ○
0501	16-12	0.033	.005 +.003 -.000 x 45°	M22520/1-01	M22520/1-12	10 lbs	Orn	Yellow
0502	16-11	0.017	.005 +.003 -.000 x 30°			6 lbs	Blue	
0503 0504	16-10	0.043	.005 +.003 -.000 x 45°	M22520/1-01	M22520/1-12	20 lbs	Red	

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

FIGURE 3. Contact and ferrule dimensions for category D only.

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CRIMP FERRULE

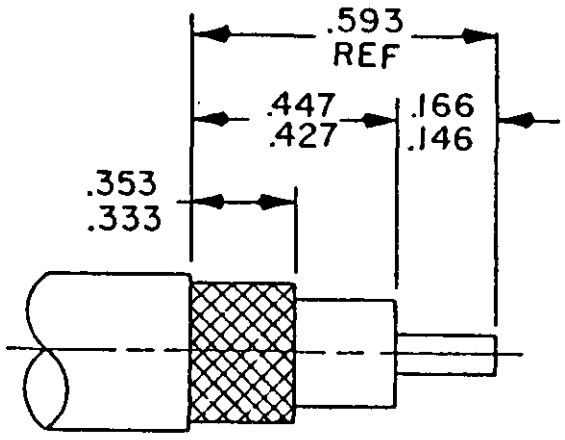
Dash No.	Ferrule No. 2/	A ±0.001	B ±0.001	Basic crimp tool 1/	Crimp die or positioner M22520/5
0501 0502	16-50	0.212	0.175	M22520/5-01	-9 Closure A -5 or -41 Closure B
0503	16-51	0.250	0.220		-5, -11, -57 Closure A or -19 Closure B
0504	16-52	0.245	0.206		

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Crimp tensile test shall be in accordance with MIL-C-39029.
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5. .003 maximum break.

FIGURE 3. Contact and ferrule dimensions for category D only - Continued.

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Inches	mm
.146	3.71
.166	4.22
.333	8.46
.353	8.97
.427	10.85
.447	11.35
.593	15.06

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 4,000 MHz.

Voltage rating:

500 volts rms, maximum working voltage at sea level.

125 volts rms, maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: 3 pounds, maximum.

Torque: 2-1/2 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque not applicable.

Mating characteristics:

Reference MIL-STD-348 and figure 2 for dimensions.

Outer contact:

Test ring ID: .319 maximum, 16 microinch finish.

Insertion force: 5 pounds, maximum when inserted a minimum of .093.

Contacts with slotted members: Shall contact a .324, minimum diameter ring within .031 of their tip ends.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B, 5,000 megohms, minimum.

Center contact retention: 6 pounds minimum axial force. Applicable to captivated-center-contact connectors only.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 500 to 4,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower; 1.30, maximum.

Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.015+.005 F (F in GHz).

Item 16: VSWR shall be less than 1.015+.005 F (F in GHz).

Second step of VSWR checkout procedure: VSWR shall be less than 1.045+.019 F (F in GHz).

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Group B inspection: VSWR shall be less than 1.1+.01 F (F in GHz).

Qualification and group C inspection: VSWR shall not exceed 1.15.

Connector durability: 500 cycles, minimum at 12 cycles per minute, maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Initial: 5 pounds, maximum.

Final: 5 pounds maximum; 1 pound, minimum.

Contact resistance: In milliohms, maximum:

	Initial	After environment
Center contact	1.5	2.0
Outer contact	.2	Not applicable
Braid to body	.1	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202, 1,500 volts rms, minimum at sea level.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition G.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be at +200°C for connectors using +200°C cables (see tables I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage: 375 volts rms, minimum.

Altitude: 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 1,000 volts rms at a frequency from 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force:

Noncrimp assemblies: 40 pounds, minimum.

Crimp assemblies:

10 pounds, minimum for cables .155 - .189 OD.

20 pounds, minimum for cables .190 - .229 OD.

30 pounds, minimum for cables .230 - .249 OD.

40 pounds, minimum for cables .250 OD and larger.

Coupling mechanism retention force: 100 pounds, minimum.

RF leakage: -55 dB minimum, tested at a frequency between 2 and 3 GHz.

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Insertion loss:

.2 dB maximum tested at 3 GHz.

Part number: M39012/16 (dash number from table I or "B" number from table III).

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
	M39012/16	M39012/16
I	-0101 -0103 -0111 -0118 -0220	-0101 -0102 -0103 -0111 -0118 -0220
II	-0102	-0102
III	B0004 B0005 B0006 B0007 B0009 B0019	B0004 B0005 B0006 B0007 B0008 B0009 B0012 B0019
IV	B0008 B0012	B0008 B0012
V	-0013 -0014 -0016 -0020 -0222	-0013 -0014 -0015 -0016 -0017 -0020 -0222
VI	-0015 -0017	-0015 -0017
VII	-0501 -0503 -0504	-0501 -0502 -0503 -0504
VIII	-0502	-0502

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

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TABLE III. Category B - Nonfield replaceable (special tools may be required).

Not for Air Force or Navy use. For OEM use only.

Dash number 1/ 2/ 3/	Applicable cable M17/ #	Dimensions	Inches (millimeters) maximum
M39012/16B0004	028-RG058*	A	1.500 (38.10)
M39012/16B0005	084-RG223*		
M39012/16B0006	111-RG303*		
M39012/16B0007	060-RG142@*		
M39012/16B0008	29-RG59*Δ 030-RG062Δ 97-RG210Δ		
M39012/16B0009	054-RG122*		
M39012/16B0012	90-RG71*Δ		
M39012/16B0019	110-RG302*@Δ		

1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ For maintenance replacements for category B, see table V.

3/ Inactive for new design.

The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and Δ.

@ Cable to be used for the +200°C temperature cycling tests.

Δ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

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TABLE IV. Cross reference of part numbers.

Preferred part number M39012/16	Substitute for part number or type designation 1/2/
-0101	UG-88/U, M39012/16-0001
-0102	UG-260/U, M39012/16-0002
-0103	UG-1033/U, M39012/16-0003
B0004 3/	M23329/3-01, M39012/16-0004
B0005 3/	M23329/3-02, M39012/16-0005
B0006 3/	M23329/3-03, M39012/16-0006
B0007 3/	M23329/3-04, M39012/16-0007
B0008 3/	M23329/3-05, M39012/16-0008
B0009 3/	M23329/3-06, M39012/16-0009
-0111	M39012/16-0011
B0012 3/	M39012/16-0012
-0013	
-0014	
-0015	
-0016	
-0017	
-0118	M39012/16-0018
B0019 3/	M39012/16-0019
-0020	
-0501	
-0502	
-0503	
-0504	
-0225	UG-959/U

1/ The superseded part number or the type designation is for cross reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M39012/10-XXXX shall be used in all cases for marking and identifying the connector.

2/ The basic type designation includes all letter versions of the specified number, e.g. UG-18/U includes UG-18 A/U, UG-18B/U, etc.

3/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

TABLE V. Maintenance replacements for category B.

Category B number* Inactive for new design	Category C dash number	Category A dash number	Category D dash number
B0004	0013	0101	0504
B0005	0014	0101	---
B0006	0013	0111	---
B0007	0014	0101	0503
B0008	0015	0102	---
B0009	0016	0103	0501
B0012	0017	0102	---
B0019	0020	0118	---

*Category B connectors are for original installation only. They will not be stocked or acquired by the Government.

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Revision letters are not used to denote changes due to the extensivness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - CR, EA, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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